

Build Your Gmail AI Assistant with n8n MCP + Claude Desktop | Step-by-Step Guide

What You'll Build Today

By the end of this workshop, you'll have Claude Desktop controlling your Gmail — sending emails, reading inbox, drafting replies — all through natural conversation!

No coding required. Just follow along.

Demo Video for reference:  MCP Workbook.mp4

Welcome, AI Builders!

Remember when managing emails felt like a full-time job?

Today, we're changing that. **Forever.**

You're about to build something that 99% of people don't even know exists — an AI that can actually *do things* in your Gmail, not just talk about it.



Time Required: 30-45 minutes

Difficulty: Beginner-friendly (seriously!)

Prerequisites: A Gmail account + n8n + Claude Desktop + willingness to try something new

Quick Context: What is MCP?

MCP = Model Context Protocol

Think of it like this:

Without MCP	With MCP
-------------	----------

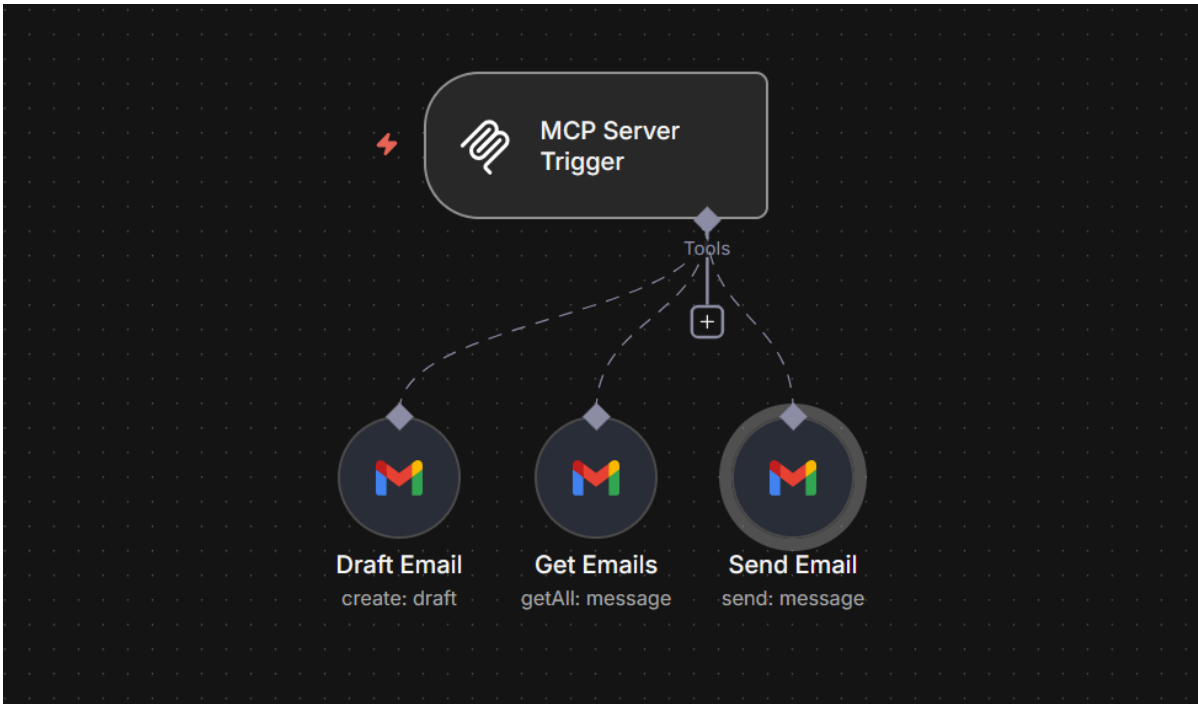
Claude can only <i>talk</i> about emails	Claude can actually <i>send</i> emails
"Here's how you would draft that..."	"Done! I've drafted your email."
Copy-paste everything manually	AI does the work for you

MCP is the bridge between AI's brain and real-world tools.

What We're Building

Your Gmail MCP Server will include:

- ✓ **Send Email** — Compose and send emails via Claude
- ✓ **Draft Email** — Create drafts for review
- ✓ **Get Emails** — Read your inbox through conversation



Prerequisites Checklist

Before we dive in, make sure you have:

- [] **n8n Account** — [Sign up free at n8n.io](https://n8n.io)
- [] **Claude Desktop App** — [Download here](#)
- [] **Gmail Account** — You've got this one 😊
- [] **Node.js installed** — [Download here](#) (LTS version)
- [] **15 minutes of focus time** — Put your phone on DND!



Pro Tip: Open this guide on one screen, and your tools on another. Makes following along 10x easier!

PART 1: Setting Up n8n

Step 1.1: Create Your n8n Workflow

▶ **Action:** Log into your n8n instance

🔗 **Link:** [n8n Cloud Login](#)

Step 1.2: Add the MCP Server Trigger

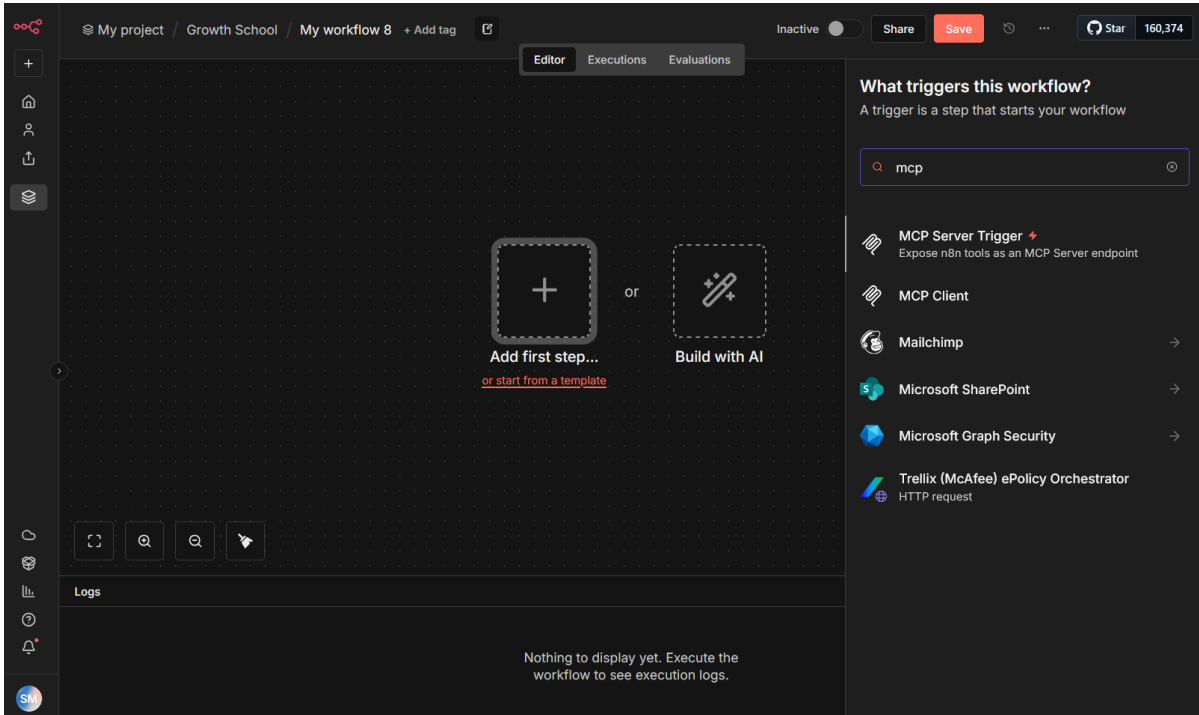
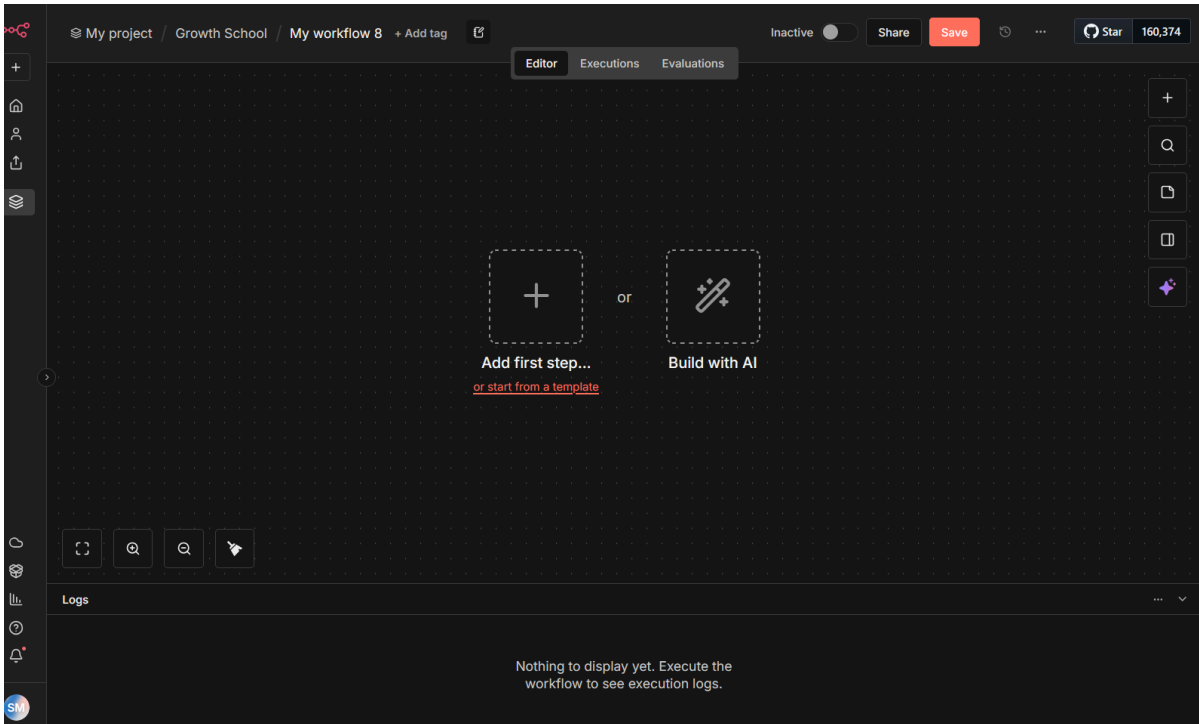
This is the magic piece that turns n8n into an MCP server!

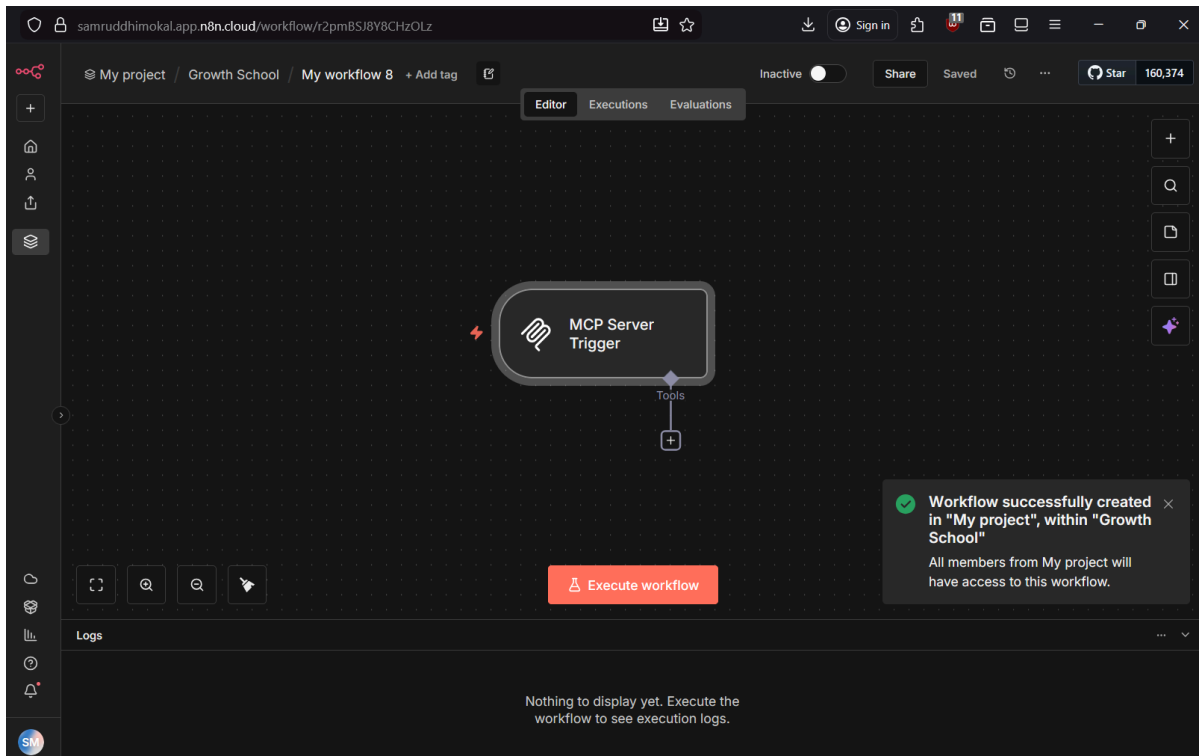
▶ **Actions:**

1. Click **" + Add first step"** or the **+** button
2. Search for **"MCP"**
3. Select **"MCP Server Trigger"**



Boom! You just created the foundation of your AI-powered email assistant!



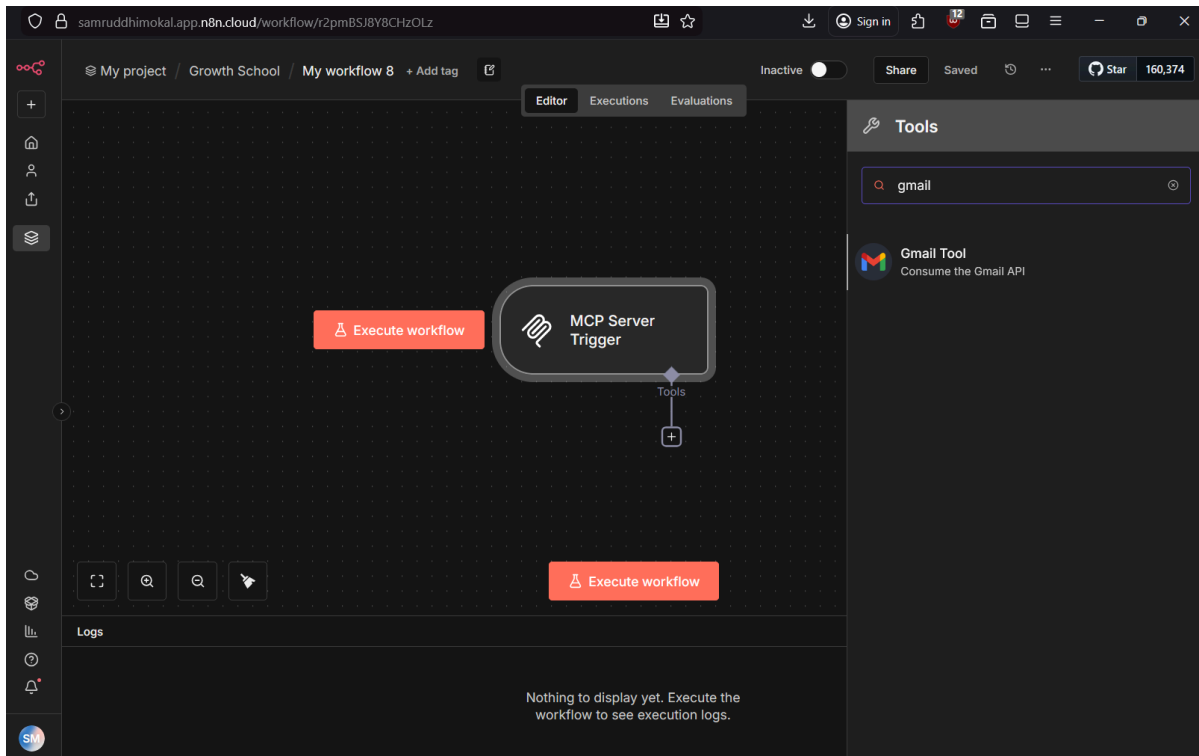


Step 1.3: Connect Your Gmail

Now let's give your MCP server email superpowers!

▶ Actions:

1. Click on **"Tools"** (the + icon below the MCP trigger)
2. Search for **"Gmail"**



3. You'll see Gmail tool — we need 3 actions of it:

Tool	What It Does	Description for AI
 Send Email	Sends new emails	"Call this to send a new email"
 Draft Email	Creates drafts	"Call this to create an email draft"
 Get Emails	Reads inbox	"Call this to retrieve emails"

Step 1.4: Authenticate Gmail

Time to connect your actual Gmail account!

▶ Actions:

1. Click on any Gmail tool node
2. Click **"Create New Credential"**
3. Sign in with your Google account
4. **Allow** the permissions (n8n needs these to manage emails)

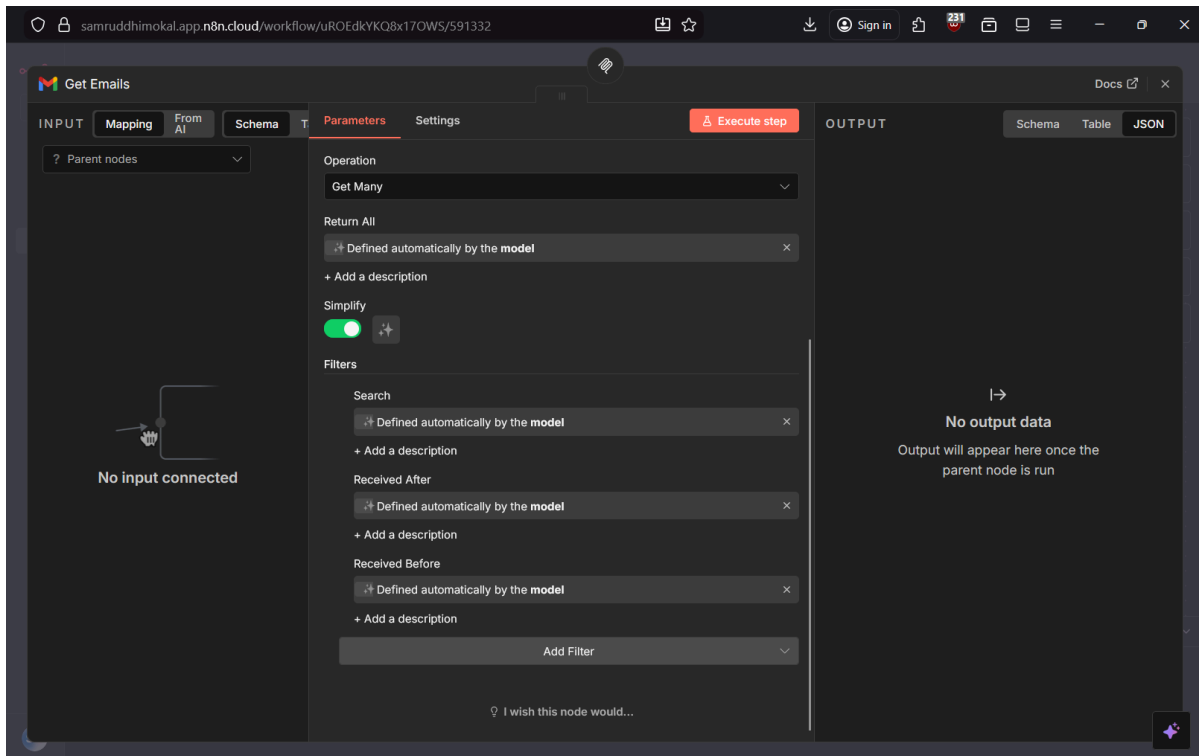
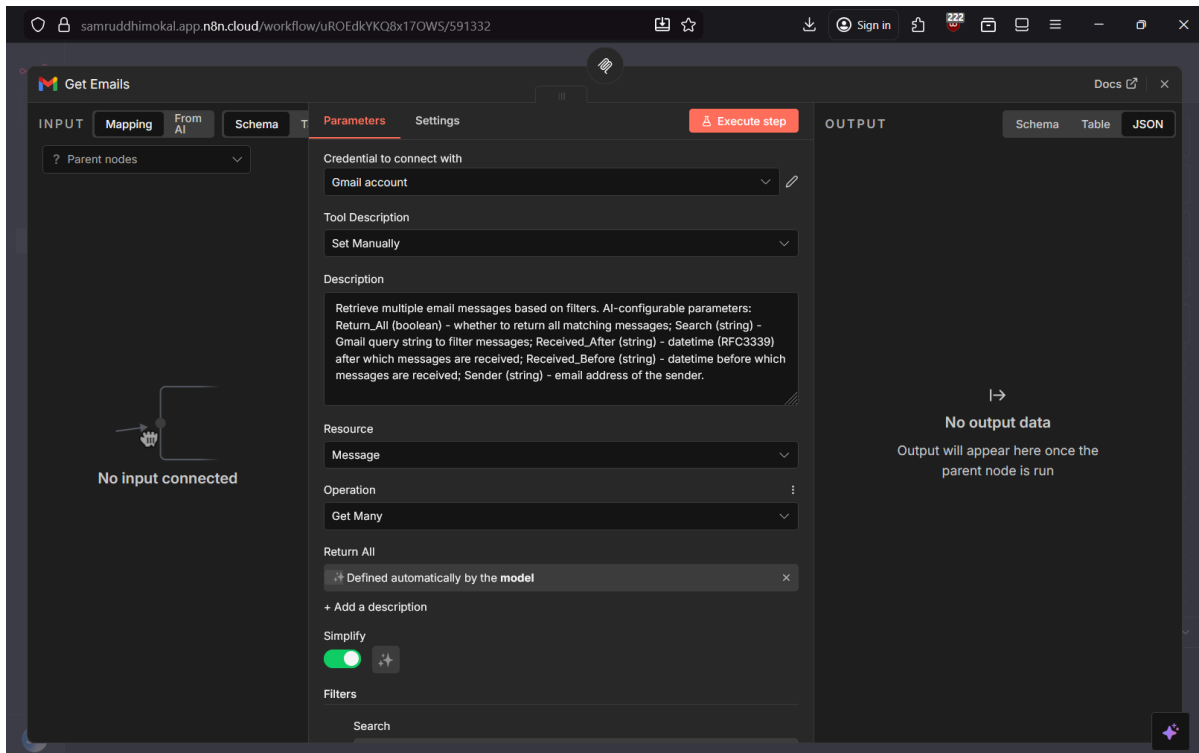
 **Security Note:** n8n only gets access to what you explicitly allow. You can revoke this anytime from your Google account settings.

Step 1.4.1: Configure Gmail Tools

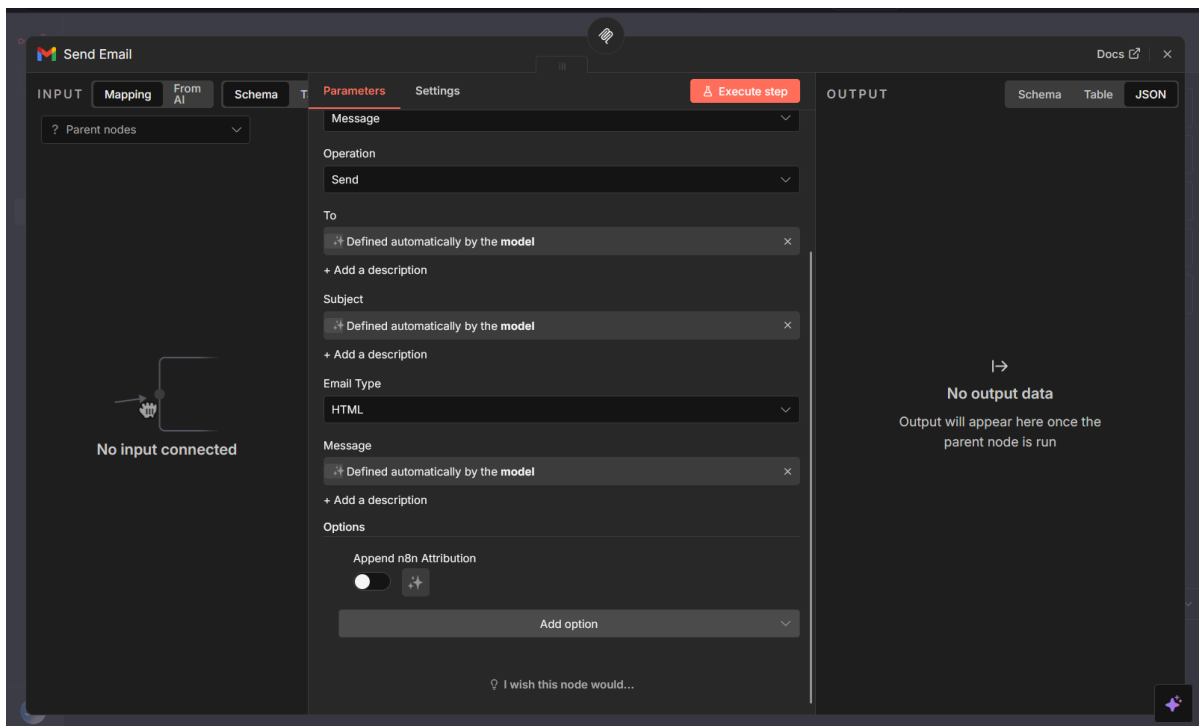
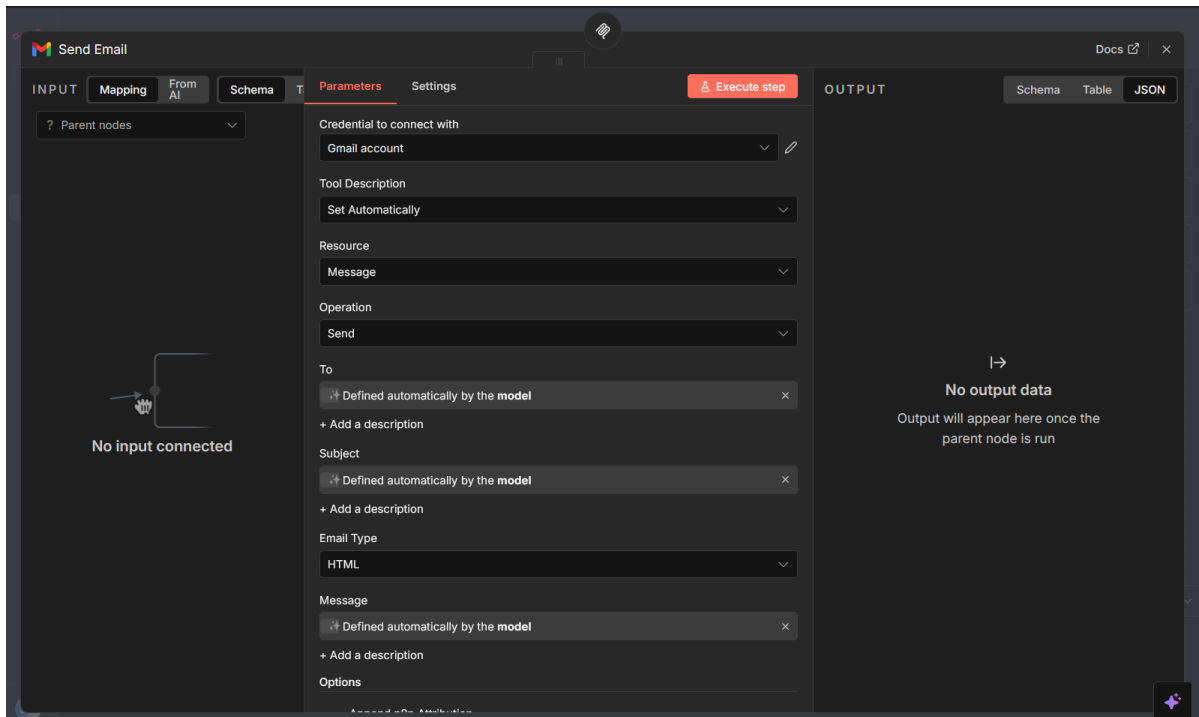
1. Configure "Get Emails". Update the Gmail Tool with the given settings in Parameters:

Copy the description from here:

Retrieve multiple email messages based on filters. AI-configurable parameters: Return_All (boolean) - whether to return all matching messages; Search (string) - Gmail query string to filter messages; Received_After (string) - datetime (RFC3339) after which messages are received; Received_Before (string) - datetime before which messages are received; Sender (string) - email address of the sender.



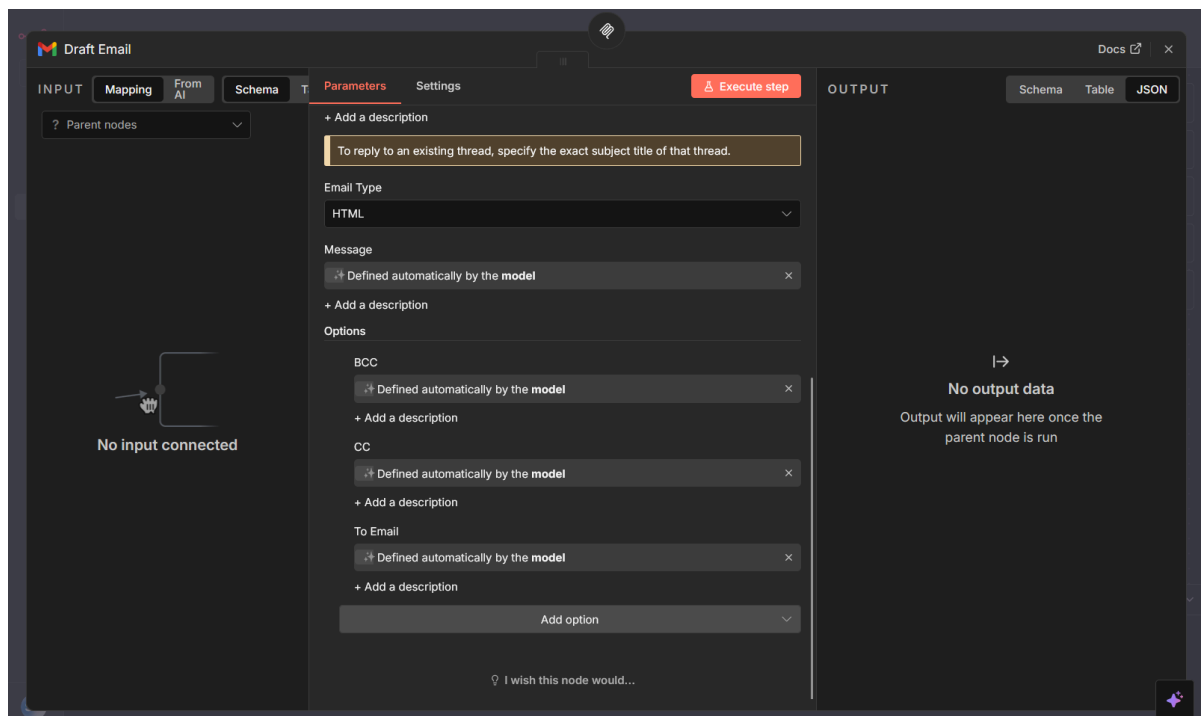
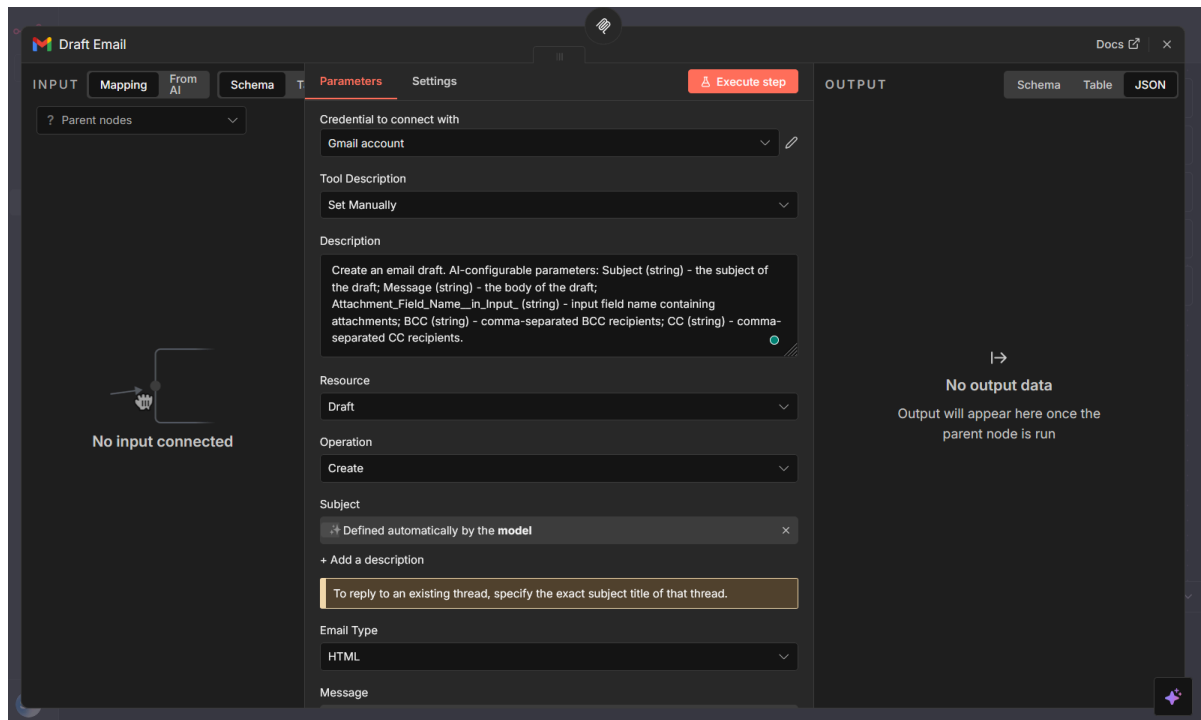
2. Copy Paste the above tool and connect to MCP server.
3. Configure "Send Emails". Update the Gmail Tool with the given settings in Parameters:



4. Copy Paste the above tool and connect to MCP server.
5. Configure “Draft Emails”. Update the Gmail Tool with the given settings in Parameters:

Copy the description from here:

Create an email draft. AI-configurable parameters: Subject (string) - the subject of the draft; Message (string) - the body of the draft; Attachment_Field_Name__in_Input_ (string) - input field name containing attachments; BCC (string) - comma-separated BCC recipients; CC (string) - comma-separated CC recipients.



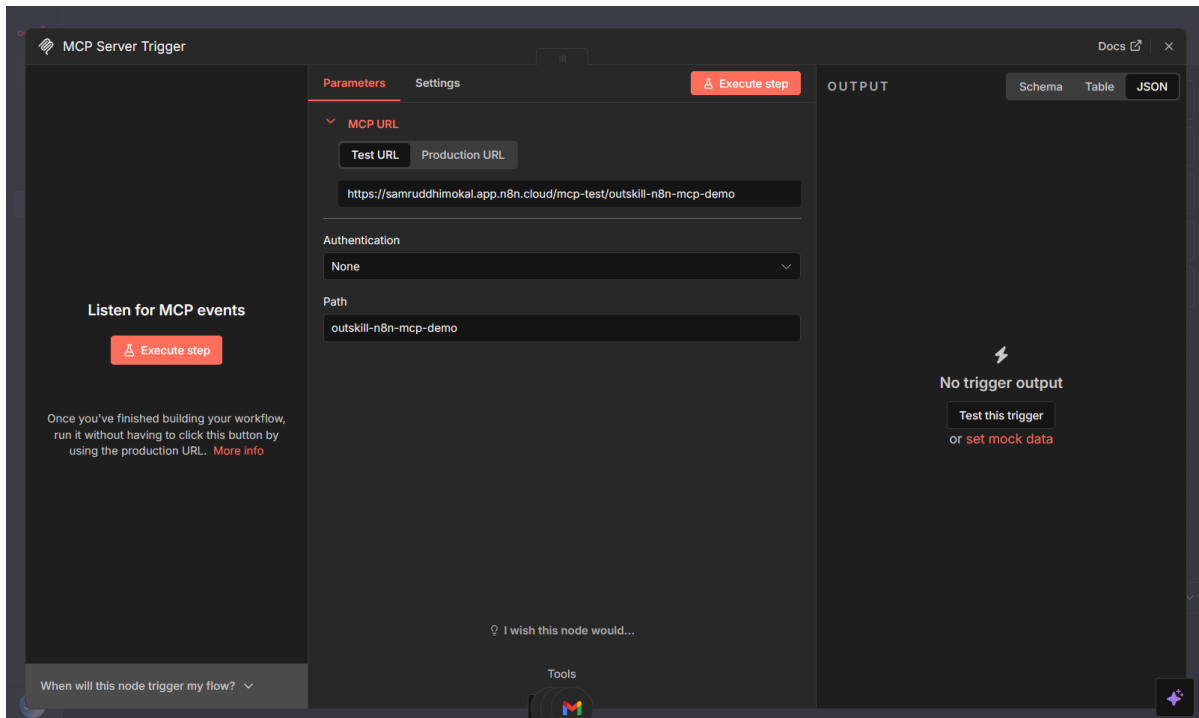
Step 1.5: Simplify Your MCP URL

▶ Actions:

1. Click on your **MCP Server Trigger** node
2. Find the **"Path"** field
3. Change it from the long random string to: **gmail-assistant** or any other custom name

Your URL should now look something like:

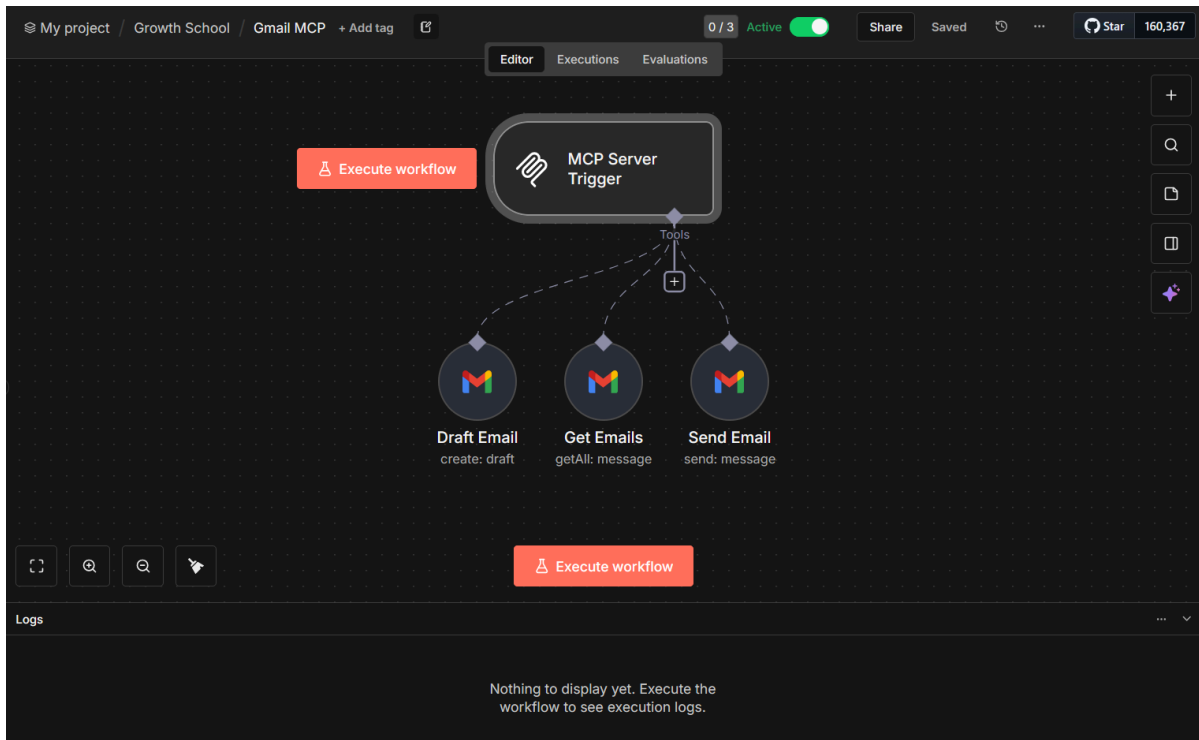
<<https://your-instance.n8n.cloud/mcp/gmail-assistant>>



Step 1.6: Save & Activate!

▶ Actions:

1. Click **"Save"** (top right)
2. Toggle **"Active"** to ON
3. **Copy your Production URL** — you'll need this!



PART 1 COMPLETE!

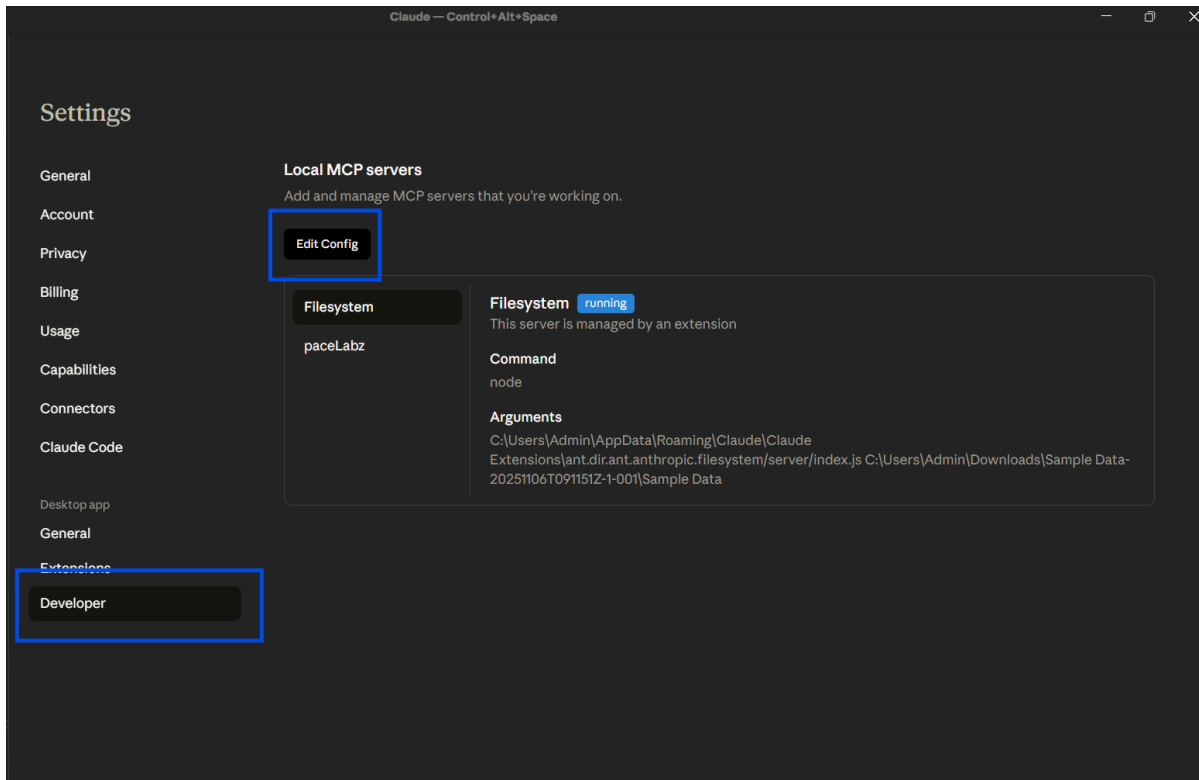
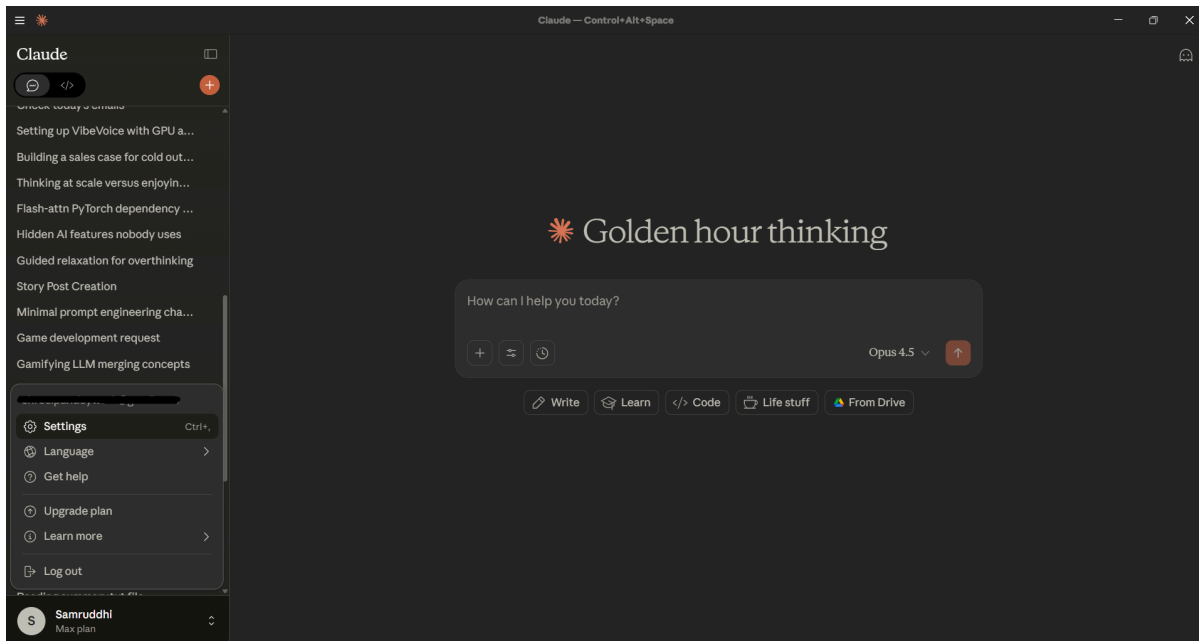
Your n8n MCP server is now live and waiting for Claude to connect! 🎉

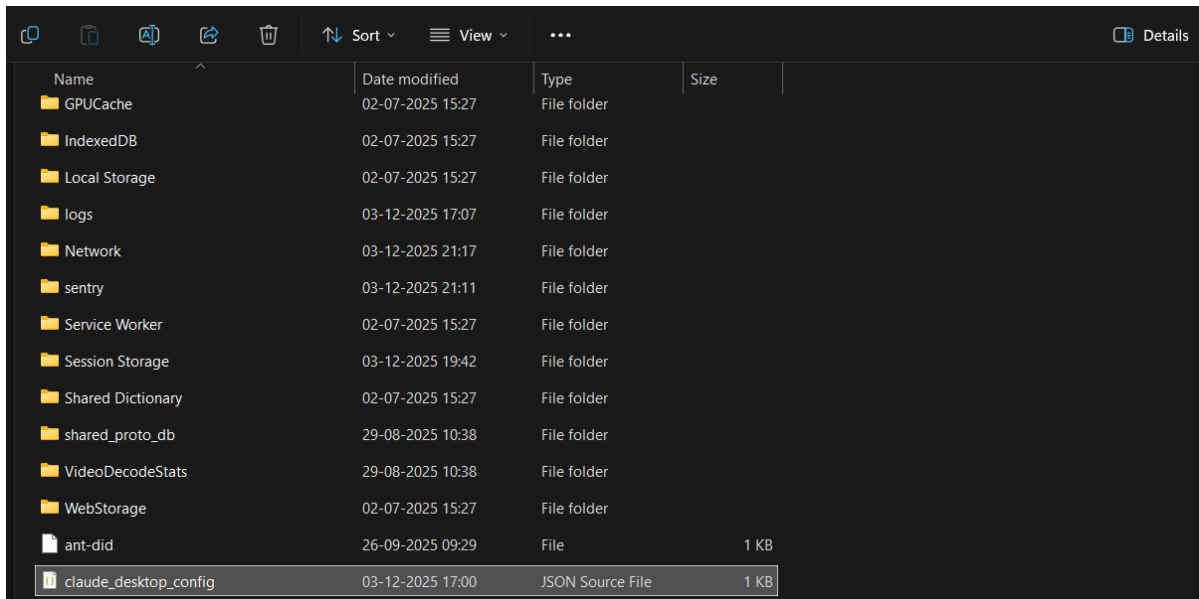
PART 2: Connecting to Claude Desktop

Step 2.1: Open Claude Desktop Settings

▶ Actions:

1. Open **Claude Desktop** app
2. Click on **Claude** menu (Mac) or **File** menu (Windows)
3. Go to **Settings** → **Developer**
4. Click **"Edit Config"**





Name	Date modified	Type	Size
GPUCache	02-07-2025 15:27	File folder	
IndexedDB	02-07-2025 15:27	File folder	
Local Storage	02-07-2025 15:27	File folder	
logs	03-12-2025 17:07	File folder	
Network	03-12-2025 21:17	File folder	
sentry	03-12-2025 21:11	File folder	
Service Worker	02-07-2025 15:27	File folder	
Session Storage	03-12-2025 19:42	File folder	
Shared Dictionary	02-07-2025 15:27	File folder	
shared_proto_db	29-08-2025 10:38	File folder	
VideoDecodeStats	29-08-2025 10:38	File folder	
WebStorage	02-07-2025 15:27	File folder	
ant-did	26-09-2025 09:29	File	1 KB
claude_desktop_config	03-12-2025 17:00	JSON Source File	1 KB



What just happened?

This opens the folder where Claude's configuration file lives. The file is called `claude_desktop_config.json`

Step 2.2: Open the Config File

Actions:

1. Find `claude_desktop_config.json`
2. Open it with any text editor (Notepad, VS Code, TextEdit)

You'll see something like this:

```
{  
}
```

Step 2.3: Add the MCP Configuration (Using NPX)

Delete everything in the file and paste this:

```
{  
  "mcpServers": {  
    "gmail-assistant": {  
      "command": "npx",
```

```

    "args": [
      "-y",
      "supergateway",
      "--sse",
      "YOUR_N8N_MCP_URL_HERE"
    ]
  }
}
}

```

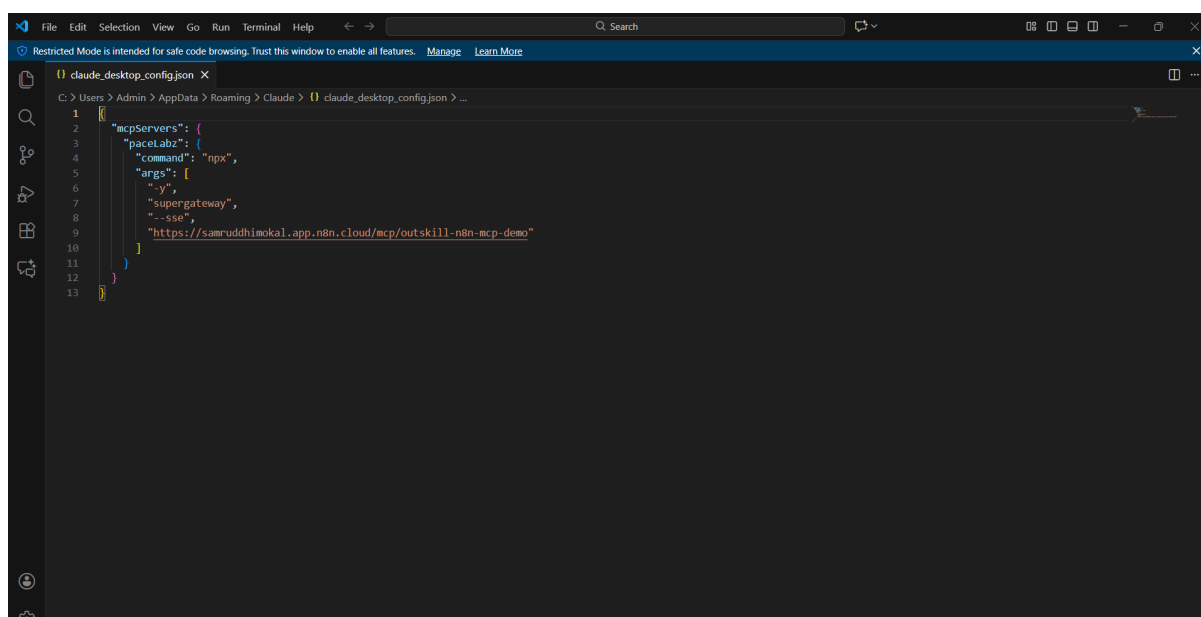
▶ **Now replace** `YOUR_N8N_MCP_URL_HERE` with your actual n8n MCP Production URL and save it!

Example:

```

{
  "mcpServers": {
    "paceLabz": {
      "command": "npx",
      "args": [
        "-y",
        "supergateway",
        "--sse",
        "<https:samruddhi.app.n8n.cloud/mcp/outskill-n8n-mcp-demo>"
      ]
    }
  }
}

```



If you have multiple MCP servers, it should look like this with MCP stacking:

```
{
  "mcpServers": {
    "gmail-assistant": {
      "command": "npx",
      "args": [
        "-y",
        "supergateway",
        "--sse",
        "https://samruddhi.cloud/mcp/outskill-n8n-mcp-demo"
      ]
    },
    "githubTools": {
      "command": "npx",
      "args": [
        "-y",
        "@anthropic/mcp-server-github",
        "--token",
        "GITHUB_TOKEN_HERE"
      ]
    },
    "localFileServer": {
      "command": "node",
      "args": [
        "/Users/you/mcp-local-files/index.js"
      ]
    }
  }
}
```

```
}  
  
}  
  
}
```



Important!

1. Use your **Production URL** from n8n (not the Test URL)
2. Don't forget the quotes around the URL!

Step 2.4: Save and Restart Claude

▶ Actions:

1. **Save** the config file (Ctrl+S / Cmd+S)
2. **Completely quit** Claude Desktop
3. **Reopen** Claude Desktop



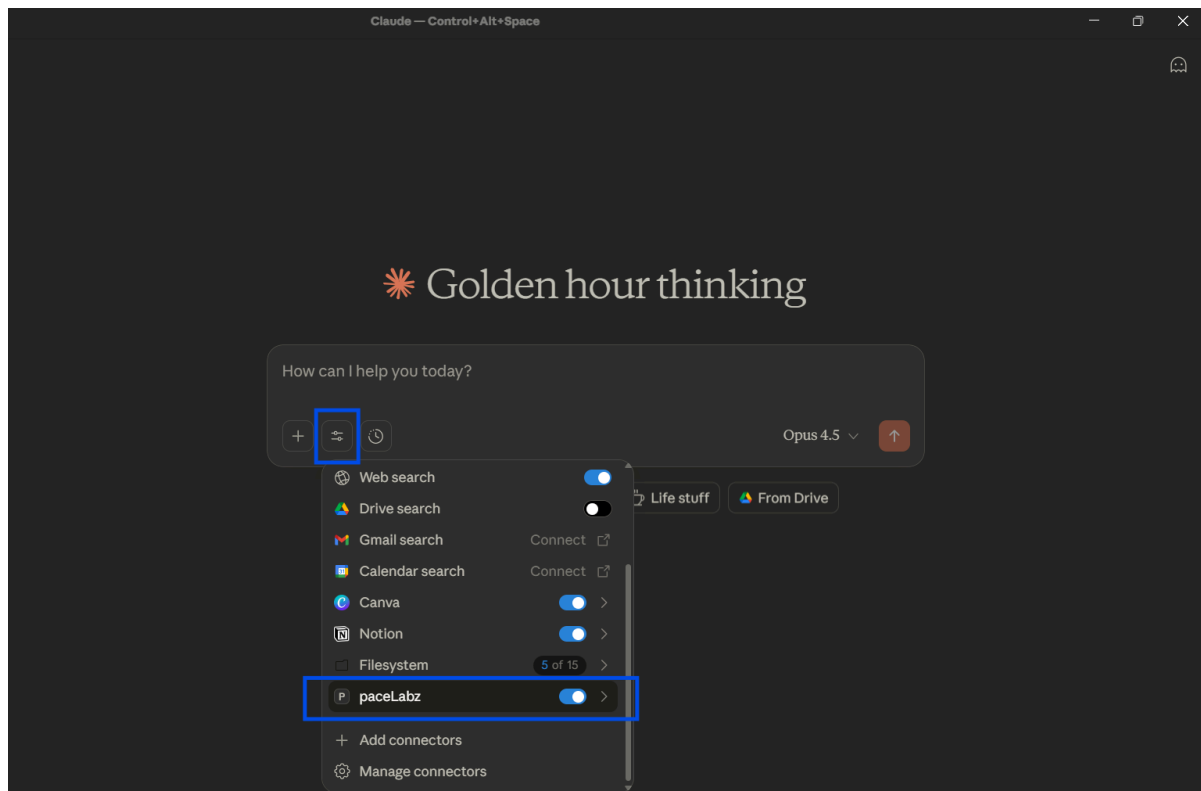
Why restart?

Claude only reads the config file when it starts up. No restart = no new tools!

Step 2.5: Verify the Connection

▶ Actions:

1. Look at the bottom of Claude's chat input
2. You should see a **tools indicator icon**
3. Click it — you should see "**gmail-assistant**"! In my case, I had named it "pacelabz"



PART 2 COMPLETE!

Claude is now connected to your Gmail through n8n! Let's test it!

PART 3: Test Your AI Email Assistant! 🧪

Test 1: Check Your Emails

▶ Type in Claude:

"Can you check my recent emails?"

Claude should ask permission to use the Gmail tool, then fetch your emails!

Test 2: Draft an Email

▶ Type in Claude:

"Draft an email to test@example.com saying I'll be 10 minutes late to our meeting"

Test 3: The Magic Moment ✨

▶ Try this:

"Check my latest emails and draft polite replies to any that need a response"

Watch Claude:

1. Read your emails
2. Analyze which need replies
3. Draft thoughtful responses

This is AI automation in action!

Troubleshooting



Problem: Claude doesn't show the gmail-assistant tool

Solutions:

1. Did you completely quit and restart Claude? (Not just close the window)
2. Check your config file for typos
3. Make sure Node.js is installed: Open terminal → type `node --version`
4. Verify your n8n workflow is **Active**



Problem: "Error connecting to MCP server"

Solutions:

1. Check your n8n URL is the **Production** URL (not Test)
2. Verify your n8n workflow is active (green toggle)



Problem: Gmail tools aren't working

Solutions:

1. Re-authenticate your Gmail in n8n
 2. Check if Google prompted you for any security approvals in your email
 3. Make sure you approved all the required permissions
-

What's Next?

You just built something incredible. But this is just the beginning!

Expand your MCP with:

-  **Google Calendar** — "Schedule a meeting with John next Tuesday"
-  **Slack** — "Post this update to the marketing channel"
-  **Google Sheets** — "Add this lead to my CRM spreadsheet"
-  **Google Drive** — "Create a new doc with these meeting notes"



The possibilities are literally endless.

Any tool that n8n can connect to, Claude can now use through MCP!

Quick Reference Card

Component	What It Is	Link
n8n	Workflow automation platform	n8n.io
MCP	Protocol connecting AI to tools	modelcontextprotocol.io
Claude Desktop	AI assistant app	claude.ai/download
Supergateway	MCP-to-SSE bridge	GitHub
Node.js	JavaScript runtime (for npx)	nodejs.org

You Did It!



Congratulations!

You've just built an AI-powered Gmail assistant that most people don't even know is possible.

This is the future of productivity. And you're ahead of the curve.
